STARLINCS

Signature-driven **T**ranscriptome **A**lignment for **R**epurposing via **LINCS**

USER MANUAL



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Table of Content

1 QUICK OVERVIEW 4

2 GETTING STARTED 6

2.1 SYSTEM REQUIREMENT 6
2.2 INSTALLATION 6
2.2.1 FOR WINDOWS 6
2.2.2 FOR WINDOWS SUBSYSTEM FOR LINUX 7
2.2.3 FOR LINUX 8
2.2.4 FOR MACOS 8

3 QUICK START GUIDE 9

3.1 LAUNCH THE COMMAND-LINE INTERFACE 9
3.2 LAUNCH THE GRAPHICAL USER INTERFACE 10
3.2.1 LAUNCH STARLINCS 10
3.2.2 SELECT REFERENCE FILE 10
3.2.3 INSERTING DIFFERENTIALLY EXPRESSED GENES 11
3.2.4 ADJUST PARAMETERS 11
3.2.5 CHOOSE MODE 11

4 REFERENCE FILE 12

4.1 BUILT-IN REFERENCE SUBSETS 12
4.2 USING A CUSTOM REFERENCE 12

5 INPUT FILES 13

6 PARAMETERS 14

6.1 --REFERENCE (GUI: SELECT ORGAN) 14
6.2 --UP (GUI: UPREGULATED GENES) 14
6.3 --DOWN (GUI: DOWNREGULATED GENES) 14
6.4 --MIN-GENE (GUI: MIN GENE) 14
6.5 --MAX-GENE (GUI: MAX GENE) 14
6.6 --SEED (GUI: SEED) 14

6.7 --nperm (GUI: Permutations) 15
6.8 --include-leading-genes 15
6.9 --output (GUI: Export CSV) 15
6.10 --mode (GUI: Mimic / Reverse) 15

7 MULTI-SAMPLE ANALYSIS 16

7.1 Automating STARLINCS in a Pipeline 16

8 UNDERSTANDING THE OUTPUT 17

9 HOW STARLINCS WORKS 18

9.1 The Core Method: GSEA Prerank 18
9.2 Why a Consensus Reference Library? 18
9.3 How the Reference Files were Built 19

10 TROUBLESHOOTING 20

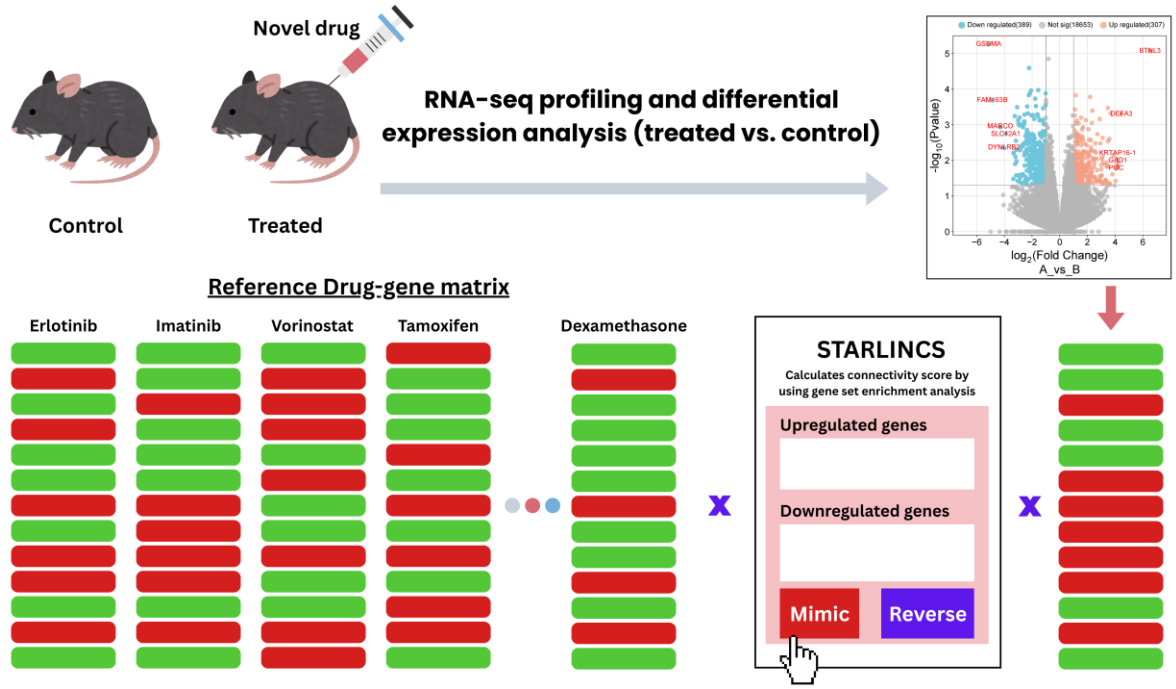
10.1 Installation Issues 20
10.1.1 "Python is not recognized" 20
10.1.2 python --version shows a version below 3.9 20
10.2 Loading Input 20
10.2.1 "Reference not found" 20
10.2.2 Gene file not recognized 20

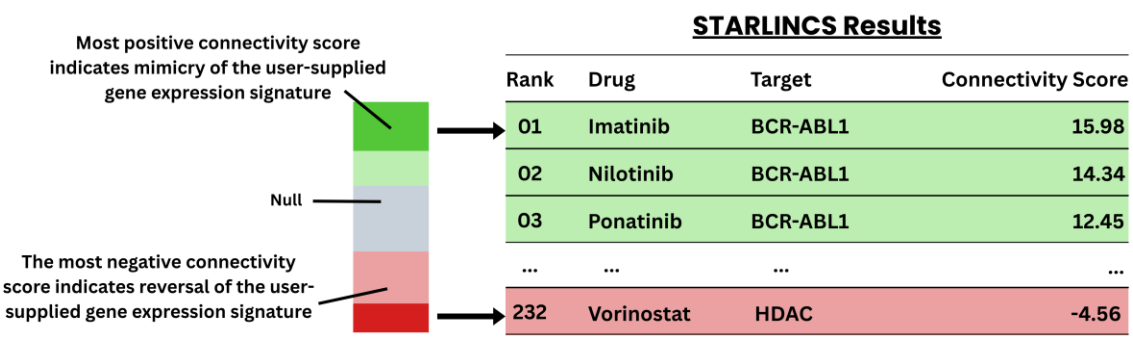
11 CONTACT 21

1 Quick Overview

STARLINCS: Signature-driven Transcriptome Alignment for Repurposing via **LINCS**

STARLINCS helps researchers identify therapeutic candidates by comparing transcriptomic gene signatures against a reference library of drug perturbation profiles. The user inputs differentially expressed genes, and the tool calculates connectivity scores for reference drugs using gene set enrichment analysis. The result is displayed as a ranking of drugs by their ability to either mimic or reverse the transcriptomic signature.





STARLINCS Results

Rank	Drug	Target	Connectivity Score
01	Imatinib	BCR-ABL1	15.98
02	Nilotinib	BCR-ABL1	14.34
03	Ponatinib	BCR-ABL1	12.45
...	...	...	...
232	Vorinostat	HDAC	-4.56

For example, a researcher performed differential gene expression analysis from a rare fibrotic lung disease (fibrotic vs. healthy lung biopsies). They can run this signature through STARLINCS to find drugs that push the transcriptome in the opposite direction from the disease state. Drugs with the most negative scores become candidates to test in a fibroblast assay for their potential to counteract fibrosis, since they are more likely to reverse its physiology as well. On the other hand, a researcher has a transcriptomic signature from cells undergoing a beneficial biological process. They query STARLINCS for drugs that mimic the same expression shift. This can surface untested compounds that reproduce the same beneficial effect, or help identify a drug's mechanism by finding what else produces a similar transcriptomic profile (useful for identifying mechanism-of-action).

The tool is available through both a graphical user interface and a command-line interface. The dual-interface framework has made it accessible to both routine use in wet labs and integration into bioinformatics pipelines at high-performance computing facilities.

2 Getting Started

STARLINCS is designed to be simple to set up and run.

2.1 System Requirement

Requirement	Details
Operating System	Windows, macOS, or Linux (cross-platform compatible)
Python Version	Python 3.9 or higher
Memory (RAM)	Minimum 4 GB recommended
Disk Space	~800 MB

2.2 Installation

2.2.1 For Windows

1. Open Command Prompt by pressing `Win` + `R` and type “cmd”.

2. Verify the Python version by running:

```
python --version
```

STARLINCS requires Python 3.9 or newer. If Python is not installed, or the installed version is older than 3.9, download and install it from python.org.

> **Important:** During Python installation, make sure to check **“Add python.exe to PATH”** before clicking **Install Now.**

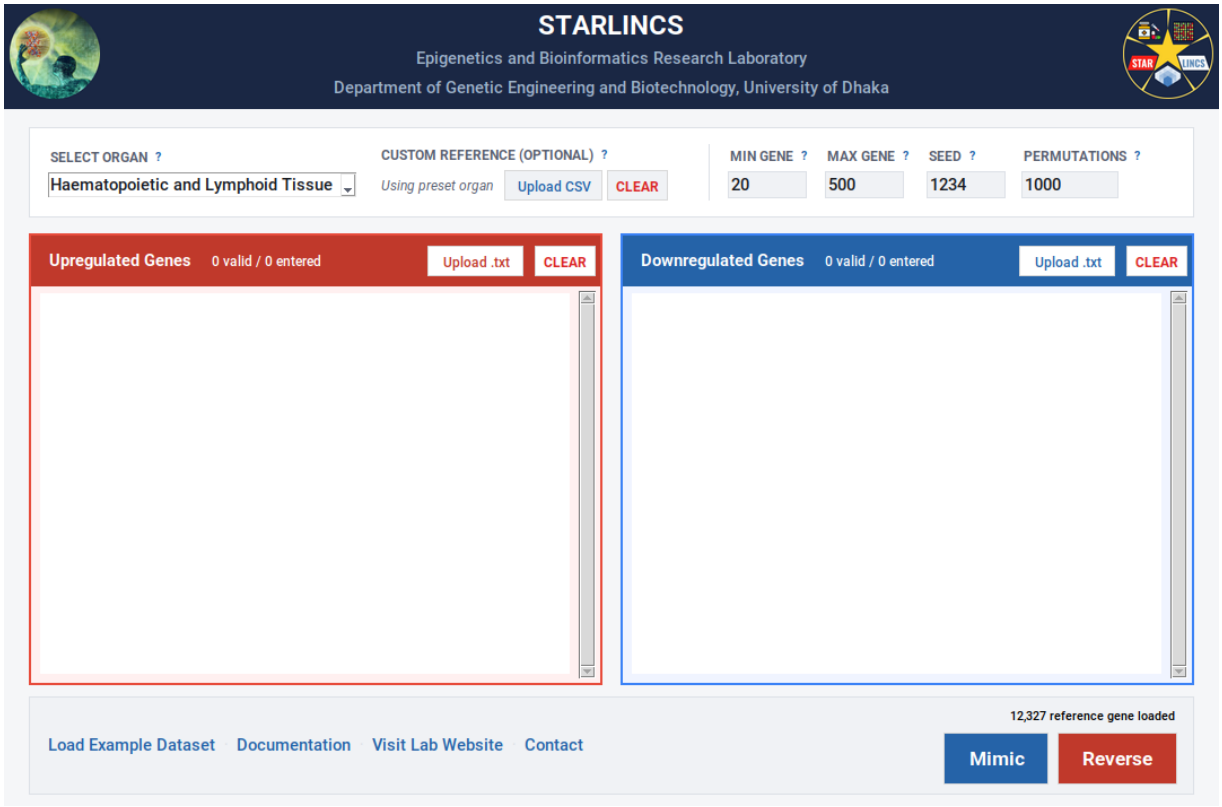
3. Download the Windows version of STARLINCS from the GitHub Repository

`Download STARLINCS for Windows`

4. Extract the .zip file to a convenient location on your computer.

5. Double-click the `setup` file to run the installer.

6. If the installation was completed successfully, enter the STARLINCS folder and double-click the **“LAUNCH_STARLINCS”** VBScript launch file. The following window will appear.



2.2.2 For Windows Subsystem for Linux

Open the WSL terminal and clone the STARLINCS repository (Option A):

```
git clone https://github.com/epigen-bioinfolab/STARLINCS
```

Alternatively, download the source code using (Option B):

```
wget https://github.com/epigen-bioinfolab/ STARLINCS/archive/refs/heads/main.zip
```

Then extract the ZIP file to a location and navigate into the extracted folder.

```
cd starlincs
```

Make the install script executable and run it:

```
chmod +x install.sh && ./install.sh
```

> **Note:** For WSL users, all commands in this section should be executed from the WSL terminal, not from Windows Command Prompt or PowerShell.

Verify that STARLINCS was installed correctly by running:

```
starlincs --version
```

2.2.3 For Linux

Open a terminal and clone the STARLINCS repository (Option A):

```
git clone https://github.com/epigen-bioinfolab/STARLINCS
```

Alternatively, download the source code using (Option B):

```
wget https://github.com/epigen-bioinfolab/ STARLINCS/archive/refs/heads/main.zip
```

Then extract the ZIP file to a location and navigate into the extracted folder.

```
cd starlincs
```

Make the install script executable and run it:

```
chmod +x install.sh && ./install.sh
```

Verify that STARLINCS was installed correctly by running:

```
starlincs --version
```

2.2.4 For MacOS

Open a terminal and clone the STARLINCS repository (Option A):

```
git clone https://github.com/epigen-bioinfolab/STARLINCS
```

Alternatively, download the source code using (Option B):

```
wget https://github.com/epigen-bioinfolab/ STARLINCS/archive/refs/heads/main.zip
```

Then extract the ZIP file to a location and navigate into the extracted folder.

```
cd starlincs
```

Make the install script executable and run it:

```
chmod +x install.sh && ./install.sh
```

Verify that STARLINCS was installed correctly by running:

```
starlincs --version
```

3 Quick Start Guide

This section walks you through running your analysis in STARLINCS, from launching the application to viewing your results.

3.1 Launch the Command-line Interface

The CLI works the same way across platforms, with minor syntax differences for multi-line commands. Before running the command below:

1. Select a valid reference name (see **Reference File**)
2. Each .txt file should contain one gene name per line. See **Input Files.**
3. Choose a mode: mimic/reverse. See **Parameters**.
   Mimic: finds drugs that mimic your signature
   Reverse: finds drugs that reverse/oppose your signature
4. Choose an output location for saving the result.

For Linux and macOS users:

Open Terminal and run:

```
starlincs run \
   --reference lung \
   --up /path/to/up_genes.txt \
   --down /path/to/down_genes.txt \
   --mode mimic \
   --output /path/to/results.csv
```

For Windows users:

Open Command Prompt (cmd) and run:

```
starlincs run ^
   --reference lung ^
   --up /path/to/up_genes.txt ^
   --down /path/to/down_genes.txt ^
   --mode mimic ^
   --output /path/to/results.csv
```

Your results will be saved to **results.csv**

For the full list of optional parameters (--min-gene, --max-gene, --nperm, --seed, --include-leading-genes), see **Parameters.**

3.2 Launch the Graphical User Interface

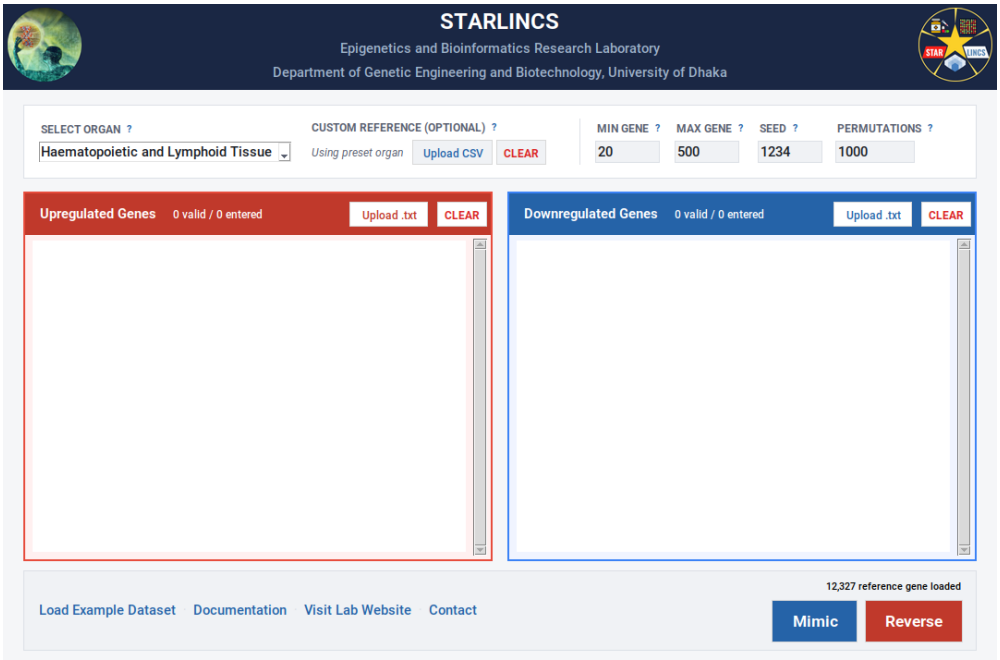
The graphical interface is the same across Windows, Linux, and macOS.

3.2.1 Launch STARLINCS

For Windows, double-click the **"LAUNCH_STARLINCS"** VBScript launch file or the STARLINCS desktop shortcut.

For Linux or macOS, go to the project folder and run:

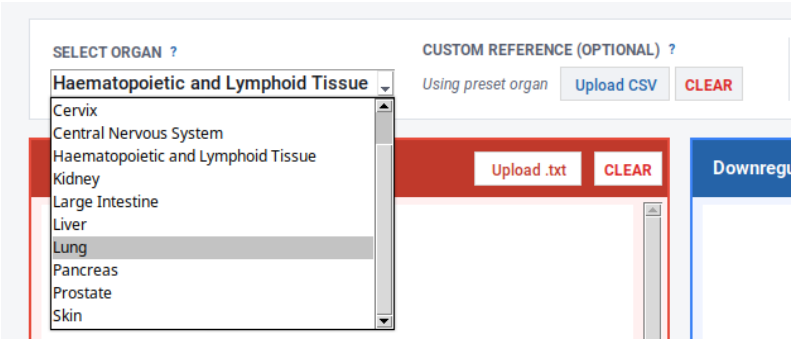
```
starlincs gui
```



3.2.2 Select Reference File

Under Select Organ, choose a built-in reference from the dropdown (e.g., Lung). To use your own reference, click Upload CSV under Custom Reference.

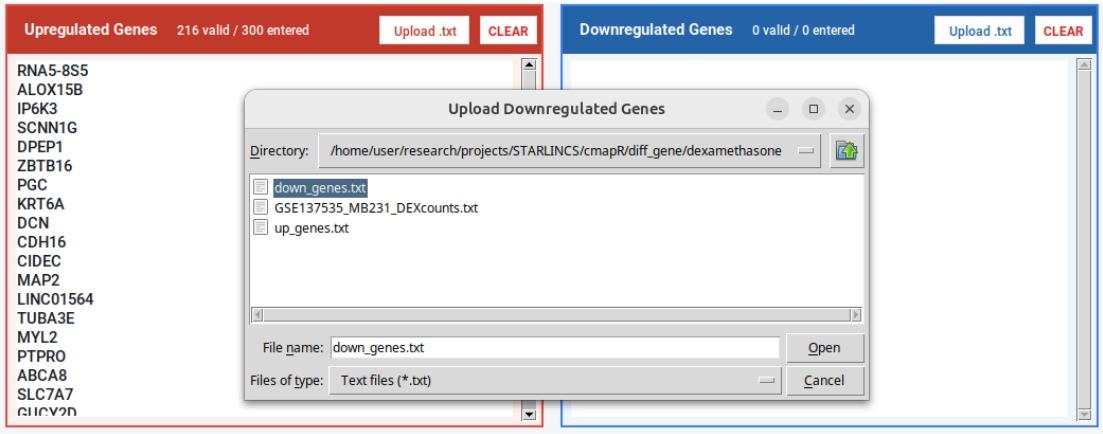
See **Reference File** for more.



3.2.3 Inserting Differentially Expressed Genes

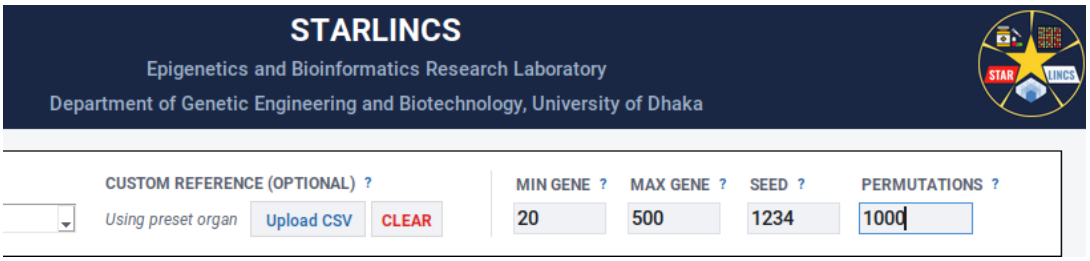
Click **Upload .txt** under Upregulated Genes or Downregulated Genes and select the gene set files. Each Text file (.txt) should contain one gene name per line.

Manual entry: Type or paste gene names directly into the input box provided for each panel. See **Input Files** for more.



3.2.4 Adjust Parameters

Adjust Min Gene, Max Gene, Seed, and Permutations if you don't want to use the defaults. See **Parameters** for details.



3.2.5 Choose mode

Determines whether STARLINCS looks for drugs with a similar (mimic) or opposing (reverse) expression pattern to your signature. See **--mode (GUI: Mimic / Reverse)** for guidance on when to use each.

Mode	Use Case
Mimic	When you are looking for compounds that can promote a desired effect, or identifying mode of action of a novel compound.
Reverse	Use this mode when you are looking for compounds that reverse a biological state. For example, identifying potential drug candidates that reverse a disease-associated gene expression signature

4 Reference File

STARLINCS compares transcriptomic gene signatures against a reference library of consensus drug perturbation profiles. You can either use one of the built-in organ-specific references or supply your own custom reference.

4.1 Built-in Reference Subsets

STARLINCS includes several pre-loaded, organ-specific reference subsets, such as:

bone	cns	pung
breast	kidney	pancreas
cervix	Large_intestine	prostate
haematopoietic_and_lymphoid_tissue	liver	skin

4.2 Using a Custom Reference

If you want to compare signature against your own dataset, you can upload a custom reference file. The file must be in CSV format. The first column (Gene) lists HGNC symbol of genes. Each column represents a ranked list, with values indicating that gene's rank for that drug. Genes are ranked from most upregulated (rank 1) to most downregulated (highest rank).

Gene	**erlotinib**	**vemurafenib**	**tamoxifen**	**…**	**sunitinib**
TP53	234	7635	2313	…	8987
ERBB2	4565	6787	23	…	11323
INS	2343	4532	2364	…	1232
…	…	…	…	…	…
…	…	…	…	…	…
HBB	4564	8775	8231	…	3454

5 Input Files

STARLINCS requires list of differentially expressed genes describing the expression changes you want to analyze. Click **Upload .txt** under Upregulated Genes or Downregulated Genes and select the gene set files. Each Text file (.txt) should contain one gene name per line.

Upregulated-genes.txt
MAP2 LINC01564 TUBA3E MYL2 PTPRO ABCA8 … … LOC101929579

Downregulated-genes.txt
CDH12 ARL14 ASB2 BCAS1 LGR5 ZNF84 … … IL11

Up- and down-regulated gene lists are typically obtained through differential gene expression analysis (e.g., disease vs. control, treated vs. untreated) using DESeq2, limma, or your preferred tool. Apply significance thresholds:

Adjusted p-value (q-value): < 0.05
Fold change (FC): commonly |log2FC| > 1.

Order each list by fold change. And save each list as a plain text file, one gene per line, with no header row.

6 Parameters

This section describes each parameter available in STARLINCS.

6.1 --reference (GUI: Select Organ)

STARLINCS compares transcriptomic gene signatures against a reference library of consensus drug perturbation profiles. You can either use one of the built-in organ-specific references or create your own custom reference. See **Reference File** for more.

* **This parameter is required.**

6.2 --up (GUI: Upregulated Genes)

File containing upregulated genes, one per line. See **Input Files.**

* **This parameter is required.**

6.3 --down (GUI: Downregulated Genes)

File containing downregulated genes, one per line. See **Input Files.**

* **This parameter is required.**

6.4 --min-gene (GUI: Min Gene)

Minimum number of genes required for a gene set to be included in the analysis.

Default: 20.

6.5 --max-gene (GUI: Max Gene)

Maximum number of genes allowed for a gene set to be included in the analysis.

Default: 500.

6.6 --seed (GUI: Seed)

Random seed used for reproducibility of permutation results.

Default: 1234.

6.7 --nperm (GUI: Permutations)

Number of permutations used to calculate statistical significance (Normalized Enrichment Score).

Default: 1000. For quick testing, values >100 are sufficient; higher values improve result stability for final analysis.

6.8 --include-leading-genes

Number of top leading-edge genes contributing most to the enrichment signal for each drug. In the graphical user interface, leading genes are displayed in the Results page. Users can export as a zip file for individual drugs.

Default: None

6.9 --output (GUI: Export CSV)

Directory or filename (.csv, .tsv, or .xls) where results will be saved.

* **This parameter is required.**

6.10 --mode (GUI: Mimic / Reverse)

Determines whether STARLINCS looks for drugs with a similar (mimic) or opposing (reverse) expression pattern to your signature.

Mode	Use Case
Mimic	To find drugs that produce a specific expression pattern. Use this mode when you are looking for compounds that can promote a desired effect, or identify the mode of action of an unknown drug
Reverse	To find drugs whose gene expression signature is opposite to the transcriptomic signature. Use this mode when you are looking for compounds that reverse a biological state. For example, identifying potential drug candidates that reverse a disease-associated gene expression signature

* **This parameter is required.**

7 Multi-Sample Analysis

Running STARLINCS manually for each gene expression signature is inefficient. It can easily be automated and integrated into a larger pipeline.

7.1 Automating STARLINCS in a Pipeline

Prepare a folder structure like this, with paired up/down gene files for each signature:

```
signatures/
├── sample1_up.txt
├── sample1_down.txt
├── sample2_up.txt
├── sample2_down.txt
├── ...
├── sample40_up.txt
└── sample40_down.txt
```

Loop through all signatures using a simple bash script:

```
#!/bin/bash
for i in $(seq 1 40); do
   starlincs run \
      --reference lung \
      --up signatures/sample${i}_up.txt \
      --down signatures/sample${i}_down.txt \
      --mode mimic \
      --output results/sample${i}_results.csv
done
```

Save this as **run_batch.sh**, make it executable, and run the following program:

```
chmod +x run_batch.sh
./run_batch.sh
```

This will process multiple signatures sequentially, saving each result as a separate .csv file in the results/ folder.

8 Understanding the Output

After running an analysis, STARLINCS generates a result listing drugs ranked by their similarity or reversal to your gene signature.

Column	Description
Drug	The name of the drug/compound from the reference library.
Target	The known molecular target(s) of the drug.
Mode of Action	Mechanism of action of the drug.
Groups	Development/approval status of the drug: approved, investigational, experimental, or vet_approved
NES_up	Normalized Enrichment Score for the up-regulated gene list
NES_dn	Normalized Enrichment Score for the down-regulated gene list
Connectivity Score	Calculated as [NES_dn − NES_up]. This combined score reflects the strength and direction of the relationship and is used to rank drugs depending on the mode selected.
Alias	An alternative name for the drug, as recorded in the LINCS database.
canonical smiles	The canonical SMILES notation representing the drug's chemical structure.
InChIKey	A standardized identifier for the drug's chemical structure, useful for cross-referencing with other chemical databases.
Leading genes	The genes contributing most to the enrichment signal for each drug. GUI: Leading genes can be exported directly from the results table. CLI: Use the --include-leading-genes N flag (e.g., --include-leading-genes 10) to obtain leading-edge genes for the top N drugs. This will be saved as a .zip file alongside output.csv

9 How STARLINCS Works

9.1 The Core Method: GSEA Prerank

The user provides a list of differentially expressed genes (up-regulated and down-regulated). STARLINCS uses the Gene Set Enrichment Analysis (GSEA) Prerank module to calculate connectivity score by comparing the gene expression signature and consensus perturbation profile of each drug in the reference file.

The gene set is "walked" along a ranked list of the drug from top to bottom, increasing a running-sum score when a matching gene is found and decreasing it otherwise. This produces a running enrichment profile whose peak deviation becomes the Enrichment Score (ES). This procedure is conceptually based on a weighted Kolmogorov–Smirnov (KS) statistic, which measures the maximum deviation between the cumulative distributions of genes inside and outside the query gene set. Raw ES values are sensitive to gene set and list size. GSEA runs permutation testing (--nperm, default 1000) to build a null distribution and calculates Normalized Enrichment Score (NES), enabling fair comparison across drugs. STARLINCS calculates this separately for up-regulated (NES_up) and down-regulated (NES_dn) genes, then combines them into the final connectivity score.

$$\text{Connectivity Score} = \text{NES}_{\text{up}} - \text{NES}_{\text{down}}$$

Results are ranked by their ability to either mimic or reverse a gene signature. A negative connectivity score indicates the drug can reverse the gene expression signature, and a positive score indicates mimicry of the signature.

9.2 Why a Consensus Reference Library?

LINCS contains over 1 million perturbation signatures of thousands of perturbagens. However, using this data directly presents two challenges:

Computational cost: querying against millions of individual signatures is computationally intensive.

Signature variability: due to inherent limitations of the L1000 assay, there is substantial variability across the multiple signatures recorded for each drug. Most existing drug-repurposing tools treat all L1000 signatures as equally reliable, without accounting for this variability.

9.3 How the Reference Files were Built

We processed LINCS L1000 signatures as follows:

Stratification by organ-of-origin: signatures are grouped into organ-specific subsets based on the tissue origin of their cell lines.

Density-based clustering: a clustering algorithm is applied within each subset to identify and extract consistent signatures, filtering out noisy perturbation profiles.

Rank aggregation: all reliable perturbation signatures associated with a given drug are aggregated into a single prototype ranked list.

10 Troubleshooting

Troubleshooting section covers the common failure points we've discussed throughout this conversation.

10.1 Installation Issues

10.1.1 "Python is not recognized"

Python is not installed, or not added to your system PATH. Reinstall Python from python.org, making sure to check "Add Python to PATH" during installation.

10.1.2 python --version shows a version below 3.9

You have an older Python version installed. Install Python 3.9 or newer from python.org, or update your existing installation.

10.2 Loading Input

10.2.1 "Reference not found"

Check that the organ name passed to --reference matches one of the built-in options (reference files are case-sensitive, e.g., lung, not Lung). Run the following command to see the full valid list or see Section 4: Reference Files.

```
Starlincs --help
```

10.2.2 Gene file not recognized

Ensure your up/down gene files are saved as plain .txt, contain one gene name per line and do not include a header row or extra columns.

11 Contact

If you run into an issue not covered in the Troubleshooting section, encounter a bug, or have a feature request, you can reach us through the following channels.

GitHub Issues

Report bugs or request features directly on the repository:
https://github.com/ epigen-bioinfolab/STARLINCS/issues

Lab Website

https://www.epigen-bioinfolab.com/

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